3-DIMENSIONAL MODELLING AND ABLATION OF LUNG TUMOUR

TEAM DETAILS:

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ABSTRACT

Lung cancer is the second widespread cancer worldwide, with almost 2.09 million new diagnoses each year. The development of the minimally invasive interventional technology recently has proven ablation therapy to be a safe and effective local treatment of cancer. Radiofrequency ablation is an effective therapy solutions for patients who have difficulty with surgery.

Computed Tomography screening is highly sensitive for the detection of lung nodules. It locates more lung cancers than chest radiography. It shows the shape, size, and position of the tumour spread in the lungs.

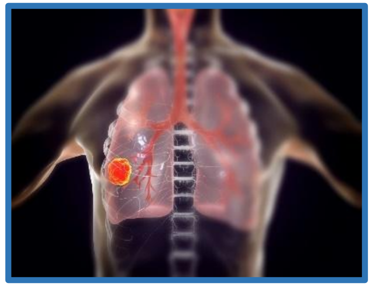


Fig 1: Lung Cancer Visualization

NEED OF THE PROJECT

Radiofrequency ablation (RFA) is an emerging minimally invasive therapy in small, localized lung cancer. Currently, no clinical trials have been published to directly compare RFA to the current standard of surgical resection. Ultimately, it will need clinical trials to evaluate oncologic outcomes involving long interval follow-up to determine survival, local control and disease progression before it becomes a reasonable and safe alternative to surgical resection.

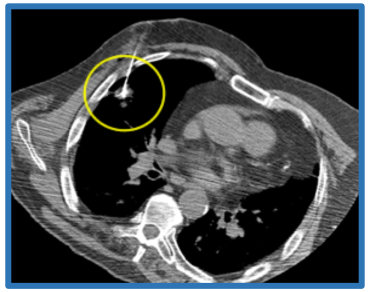


Fig 2: RF ablation of lung tumour

OBJECTIVE OF THE PROJECT

The project aims at the following:

- Segmentation of the tumour using a developed algorithm
- Segmentation of the tumour using various software such as Materialize MIMICS and 3D Slicer
- Ablation of the various tumour’s using Radiofrequency Ablation and analysis

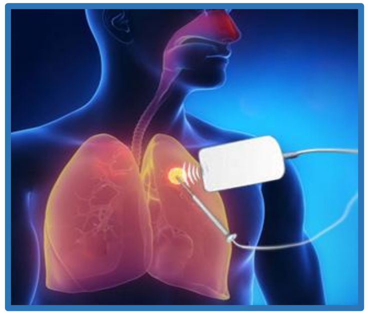


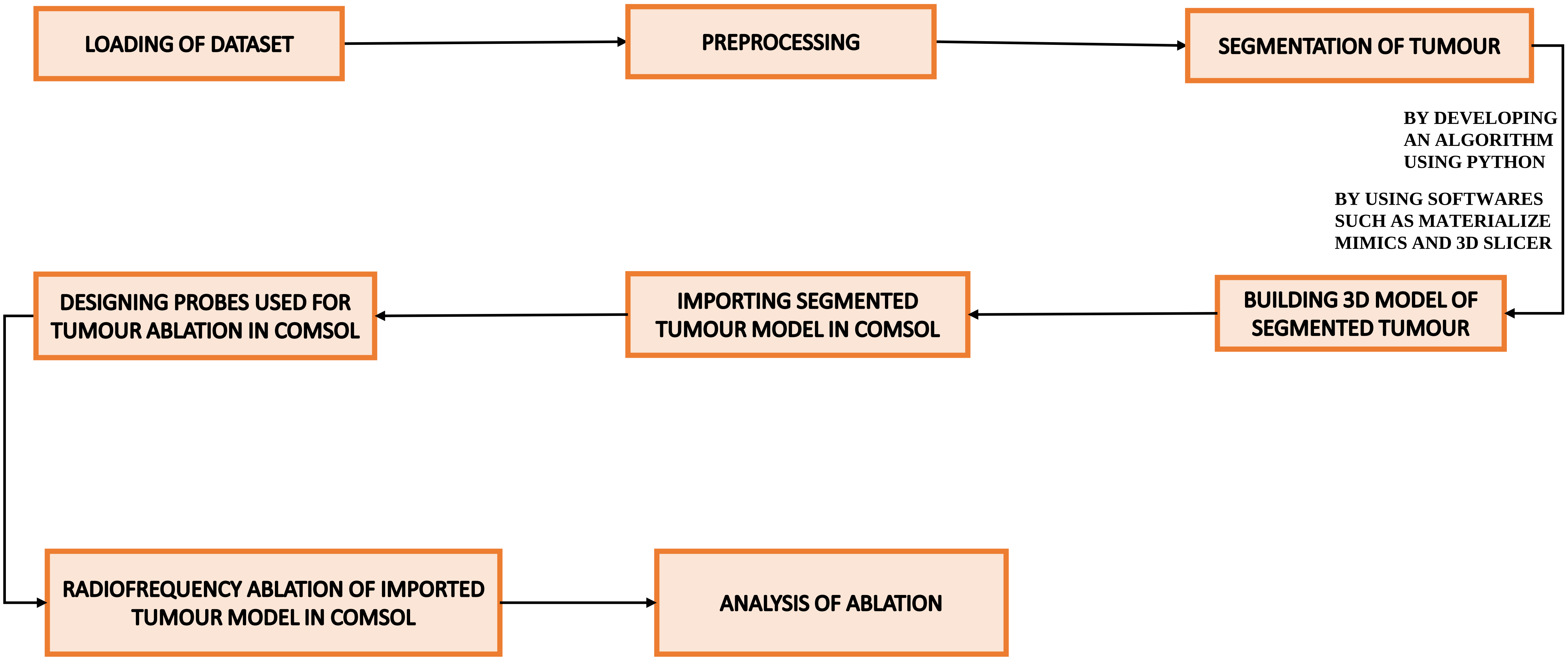
Fig 3: MW ablation of lung tumour

LITERATURE SURVEY

Title of the paper	Published journal and year	Author	Inferences
Pre - clinical modeling and simulation of radiofrequency ablation in human lung tumor	International Journal of Mechanical and Production Engineering Research and Development (IJMPERD) ISSN (P): 2249-6890; ISSN (E): 2249-8001 Vol. 7, Issue 4, Aug 2017, 77-88	Gurwinder Singh and Mahakdeep Singh	The effective tumor ablation essentially depends upon, exposure time and on voltage range. The study further evaluates the range of input voltage requirement, for attaining safe ablation of tumor tissue without damaging healthy tissue during temperature controlled RFA of lung tumor(i.e. 24-28V).
Computational modeling of gold nanoparticle in lung cancer photothermal therapy	Journal of Clinical Engineering, 44(4), pp.157-164, 2019.	Ibtihag Yahya and Megdi Eltayeb	This potential of precious metals such as AuNPs in the treatment of cancerous cells is simulated using COMSOL Multiphysics to model heat transfer in lung tumor. The study also shows how to harness the gold power and properties in killing cancer cells without harming the healthy ones. Another significant finding is that both PDE (diffusion equation) and heat transfer modules can model the thermal activation of the AuNP and predict the thermal profile inside the tumor successfully.
Temperature-controlled radiofrequency ablation of different tissues using two-compartment models	International journal of hyperthermia, vol. 33, no. 2, 122–134, 2017	Sundeep Singh and Ramjee Repaka	The ablation volume predicted by the damage front and conventional isotherms are not the same. The optimal treatment time for complete tumour ablation can play a critical role in minimising the damage to the surrounding healthy tissue and ensuring safe and risk free application of RFA. The obtained results emphasise the need for developing organ-specific clinical protocols and systems during RFA of tumour.

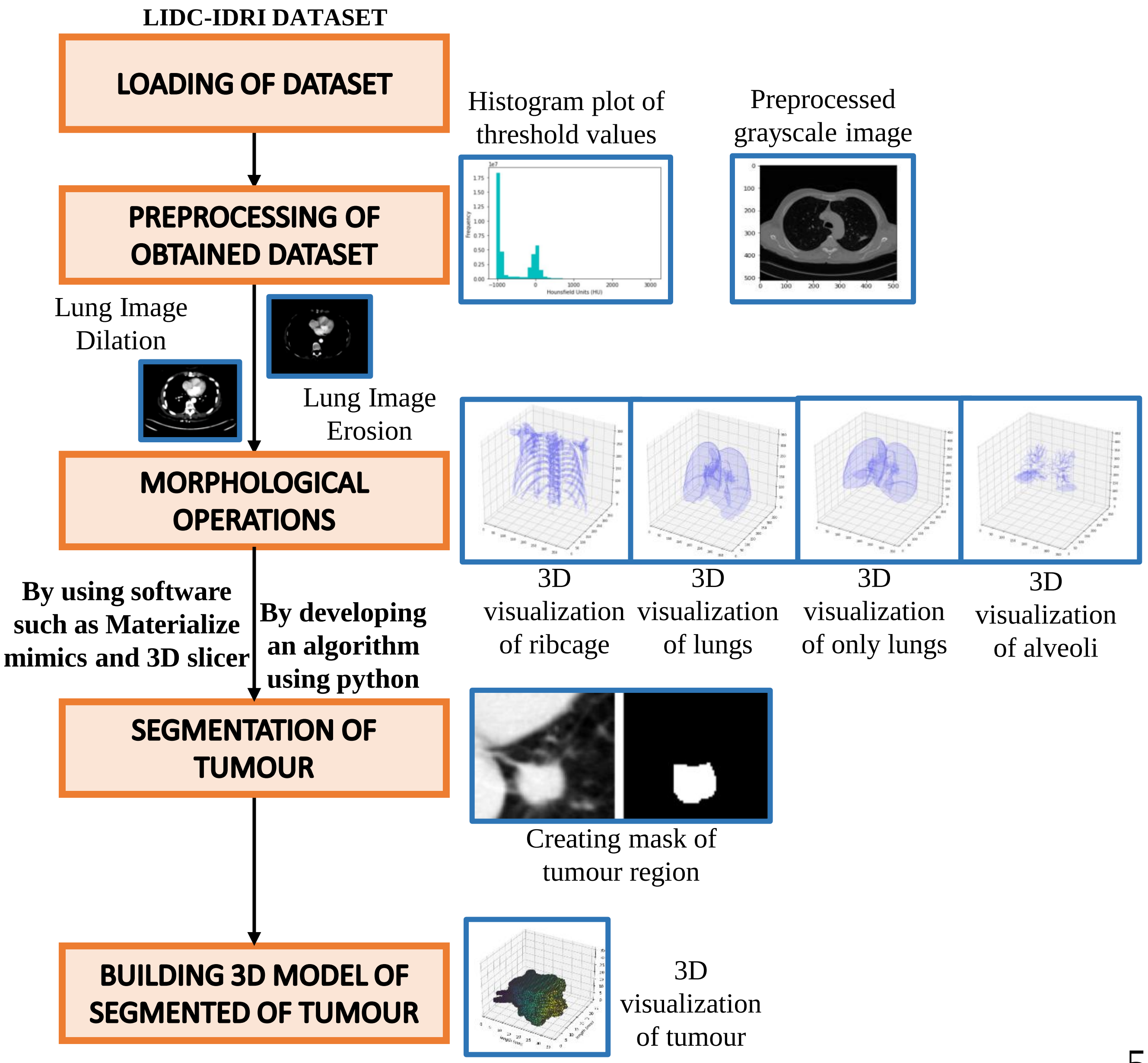
Title of the paper	Published journal and year	Author	Inferences
Thermal damage modeling analysis and validation during treatment of tissue tumors	International Journal of Pharma Medicine and Biological Sciences Vol. 6, No. 4, October 2017	Mhamed Nour, Aziz Oukaira, Mohammed Bougataya, and Ahmed Lakhssassi	The effective tumor ablation essentially depends upon, exposure time and on voltage range. The study further evaluates the range of input voltage requirement, for attaining safe ablation of tumor tissue without damaging healthy tissue during temperature controlled RFA of lung tumor. The results from the current study may be useful for the clinical practitioners, by providing them guidelines and it could make the RFA more effective and reliable.
Simulation study of microwave heating with nanoparticle diffusion for tumor ablation	International Journal Bioautomation, 23(4), p.421, 2019	Kazi Mahdi Mahmud and MD. Maruf Hossain Shuvo	The study focuses on optimizing the surrounding healthy tissue damage by diffusing magnetic nanoparticles (NPs) in the tumor region. The performance of microwave heating with and without nanoparticle diffusion is compared using the performance parameters. Simulation results show that the heating procedure coupled with ferromagnetic nanoparticle diffusion has approximately 14% less damage to the healthy tissue. The study also shows that 433/915 MHz frequency value provides 3% less damage to the healthy tissue than 2.45 GHz. Analyzing the performance of different nanoparticle we found that the ferromagnetic nanoparticle provides 15% and 5% less damage to the healthy tissue than gold and manganese iron oxide NPs respectively. The obtained result was also verified for kidney, breast, and lung tumor ablations to confirm the findings.

PROJECT WORKFLOW



METHODOLOGY

- The **LIDC-IDRI** dataset has been used for this project. Nine patients dataset has been selected for processing in this project.
- **Morphological operations** rely only on the relative ordering of pixel values.
- **Dilation** adds a layer of pixels to both the inner and outer boundaries of regions. It acts like local maximum filter. It makes objects more visible and fills in small holes in objects.
- **Erosion** is the straight opposite to the dilation. It acts like a local minimum filter. It removes islands and small objects so that only substantive objects remain.
- **Pulmonary nodules** can be diagnosed as cancer-based on the **characteristics of shape**, for example, sphericity and speculation, and composition of interior structures, like fluid, calcification, and fat.
- In the **preprocessing steps**, we obtained the grayscale image for the Patient datasets. The histogram is visualized, from which Hounsfield Unit of the image is inferred.
- For the 3-dimensional visualization of lung and its nodules, **connected component analysis** method is used.



RESULTS AND DISCUSSION

3D RECONSTRUCTION OF TUMOUR USING PYTHON ALGORTIHM

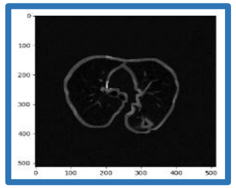


Fig 4: Boundary Segmented Image

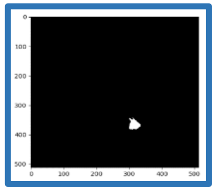


Fig 5: Tumour Masked Image

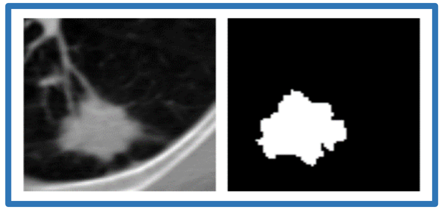


Fig 6: Visualization of the masked tumour image

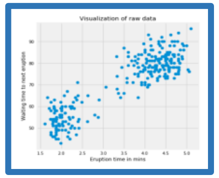


Fig 7: Visualization of raw data

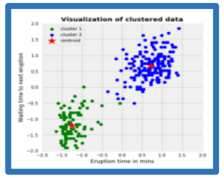


Fig 8: Visualization of clustered data

- In the preprocessing steps, the boundaries of the lungs and the tumour are extracted for easier visualization.
- K means clustering algorithm is an iterative algorithm that tries to partition the dataset into K pre-defined distinct non-overlapping subgroups (clusters) where each data point belongs to only one group.
- It tries to make the intra-cluster data points as similar as possible while also keeping the clusters as different (far) as possible.
- It assigns data points to a cluster such that the sum of the squared distance between the data points and the cluster’s centroid (arithmetic mean of all the data points that belong to that cluster) is at the minimum.
- The K-means clustering algorithm works as follows:
    - ➢ Specify number of clusters K.
    - ➢ Initialize centroids by first shuffling the dataset and then randomly selecting K data points for the centroids without replacement.
    - ➢ Keep iterating until there is no change to the centroids. i.e assignment of data points to clusters isn’t changing.
- The approach k means follows to solve the problem is called Expectation-Maximization.
- The E-step is assigning the data points to the closest cluster.
- The M-step is computing the centroid of each cluster.

- The objective function is:

$$J = \sum_{i=1}^{m} \sum_{k=1}^{K} w_{ik} \left\| x^i - \mu_k \right\|^2 \qquad (1)$$

where $w_{ik}=1$ for data point x_i if it belongs to cluster k; otherwise, $w_{ik}=0$. Also, μ_k is the centroid of x_i*'s* cluster.

- To minimize J w.r.t. w_{ik} and treat μ_k fixed and vice versa, we differentiate J w.r.t. w_{ik} first and update cluster assignments (E-step).
- Similarly, differentiate J w.r.t. μ_k and recompute the centroids after the cluster assignments from previous step (M-step). Therefore, E-step is:

$$\frac{\partial J}{\partial w_{ik}} = \sum_{i=1}^{m} \sum_{k=1}^{K} \left\| x^i - \mu_k \right\|^2 \qquad (2)$$

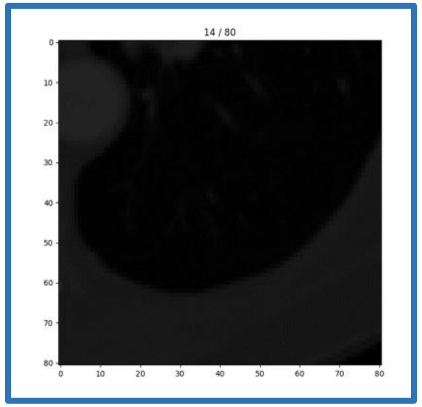
$$\Rightarrow w_{ik} = \begin{cases} 1 & , if\ k = argmin_j \left\| x^i - \mu_k \right\|^2 \\ 0, & otherwise \end{cases}$$

- In other words, assign the data point x^i to the closest cluster judged by its sum of squared distance from cluster centroid.
- And M-step is:

$$\frac{\partial J}{\partial \mu_k} = 2 \sum_{i=1}^{m} \left(x^i - \mu_k \right) = 0 \qquad (3)$$

$$\Rightarrow \mu_k = \frac{\sum_{i=1}^{m} w_{ik} x^i}{\sum_{i=1}^{m} w_{ik}}$$

- It translates to recomputing the centroid of each cluster to reflect the new assignments.
- The tumour mask is created using the K-means clustering algorithm.
- The mask image is created for each slice of the DICOM stack where the tumour is present. These images are saved as a stack of numpy arrays.
- For accuracy, the annotations of the tumour, a boundary is created across the tumour. If more than one annotation file is present, a 50% consensus boundary is created.



Masked Tumour Slices

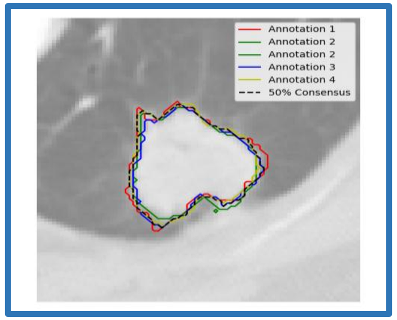
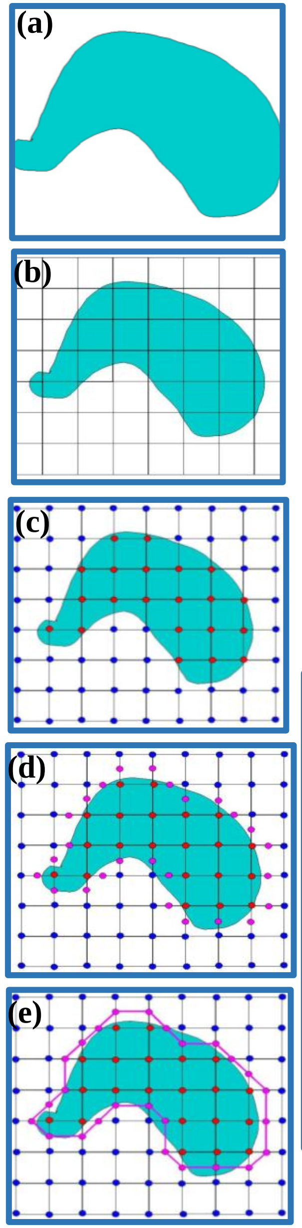
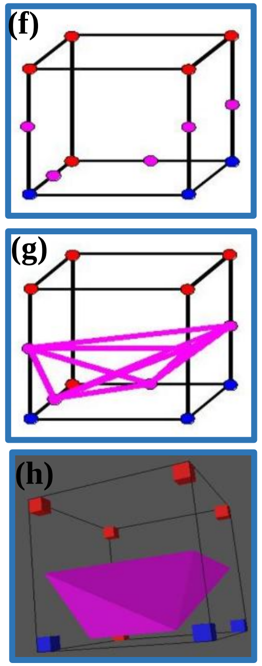


Fig 8: Annotation Boundary Image

Fig 9: Marching Cubes Algorithm





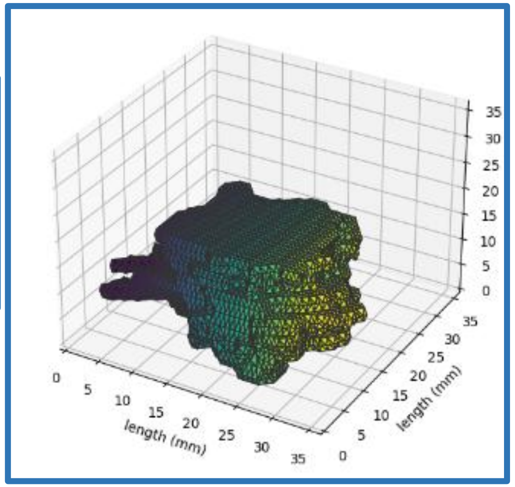
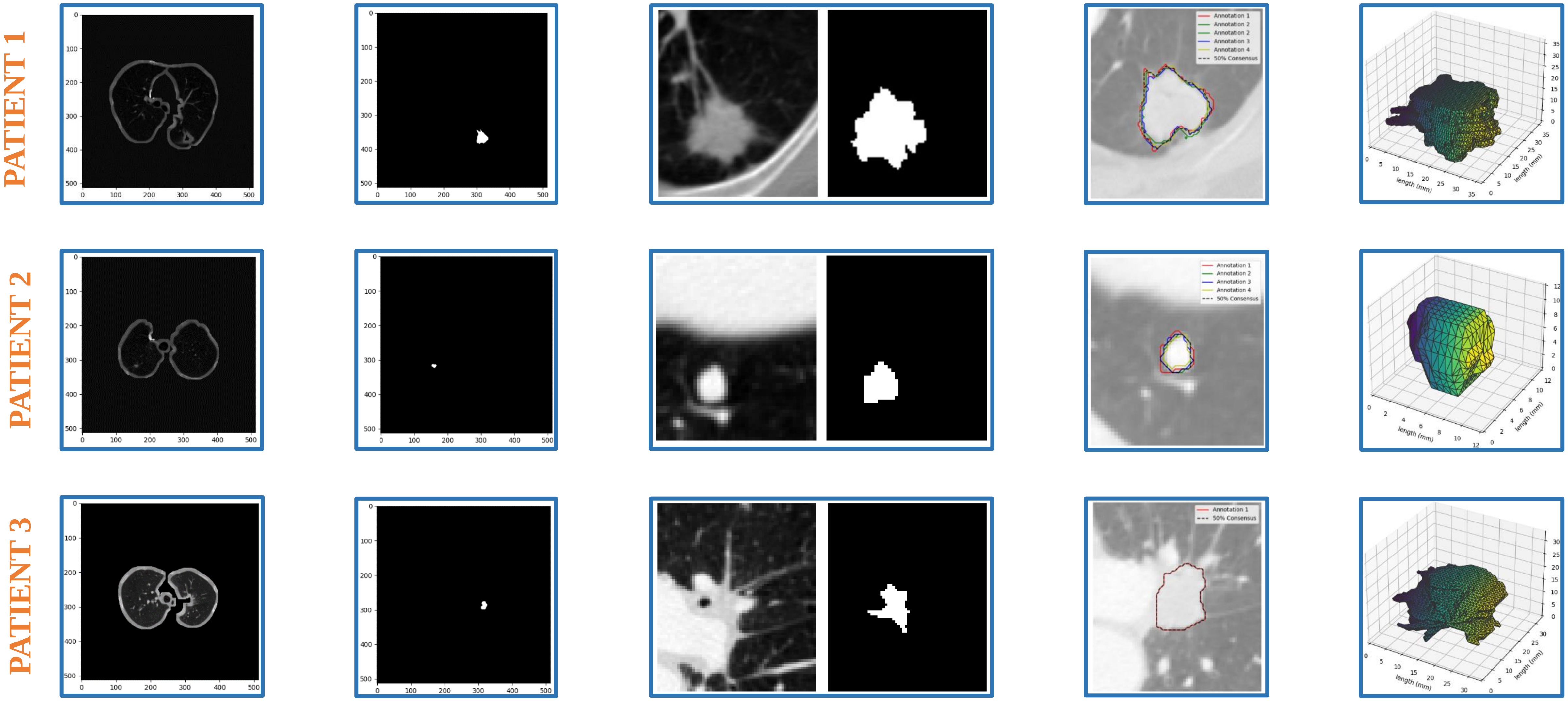


Fig 10: 3D Reconstructed Tumour

- Using the Marching Cubes algorithm, the 3-dimensional structure of the tumour is reconstructed from the stack of numpy arrays.
- Figure 9(b) depicts Voxelization.
- Then, the vertices of the square that lies inside and outside the region of interest are labelled as shown in Figure 9(c).
- The points of intersections of the boundaries with the edges as calculated by the algorithm is created as shown in Figure 9(d) and these dots are connected as shown in Figure 9(e) for an approximation of the original surface.
- In 3D, these slices are stacked with their slice thickness known. In our dataset, the slice thickness was resampled to 3mm.
- A cube is created connecting the vertices of the square present in each slice (Figure 9(f)). The dots present on the sides of the cubes (Figure 9(g)) are connected to form a triangular surface like mesh (Figure 9(h)).



(I) Boundary Segmented Image

(II) Tumour Masked Image

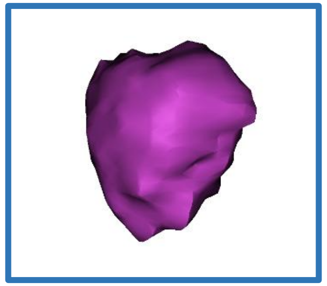
(III) Visualization of the masked tumour image

(IV) Annotation Boundary Image

(V) 3D Reconstructed Tumour

3D RECONSTRUCTION OF TUMOUR USING MATERIALIZE MIMICS SOFTWARE

- Materialize MIMICS software is used to three-dimensionally reconstruct the lung tumour for future ablation.
- Three dimensional visualization helps us to detect various clinical conditions with the shape and size of the lungs.
- The tumour is segmented, visualized and analyzed.
- At first, MIMICS imports 2D stacked images such as CT, MRI or Microscopy data in a wide variety of formats, far beyond DICOM.
- MIMICS displays the image data in several ways, each providing unique information. MIMICS include several visualization functions such as contrast enhancement, panning, zooming and rotating of the 3D view.
- In MIMICS, segmentation masks are used to highlight Regions of Interest (ROI). MIMICS enables one to define and process images with up to 30 colored segmentation masks. To create and modify these masks, the following functions are used:
    - Thresholding
    - Region Growing
    - 3D rendering information



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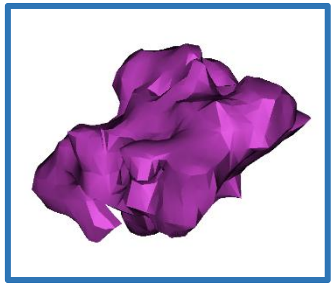

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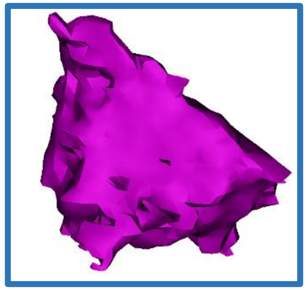
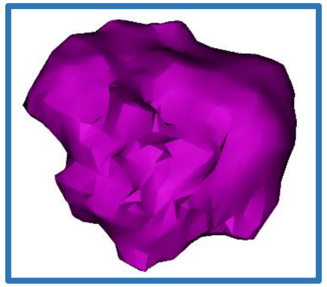


PATIENT 3



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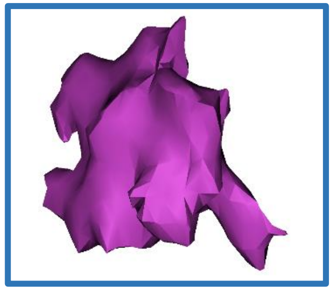

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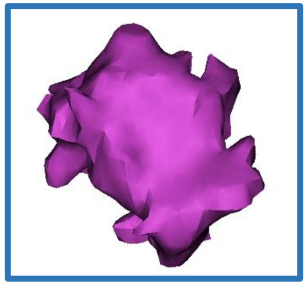

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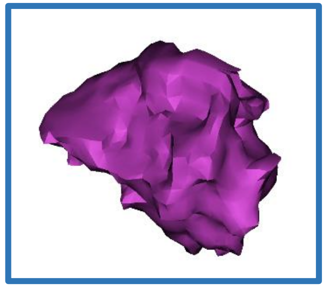

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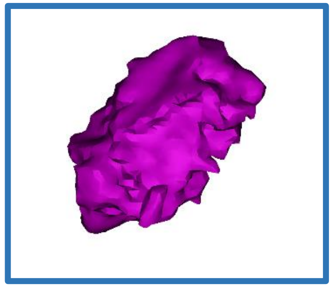

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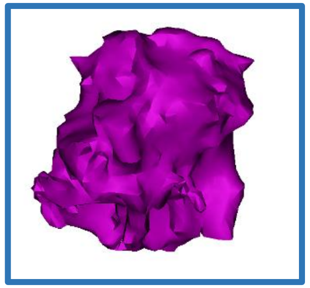


PATIENT 9

Fig 11: The different views present in MIMICS software

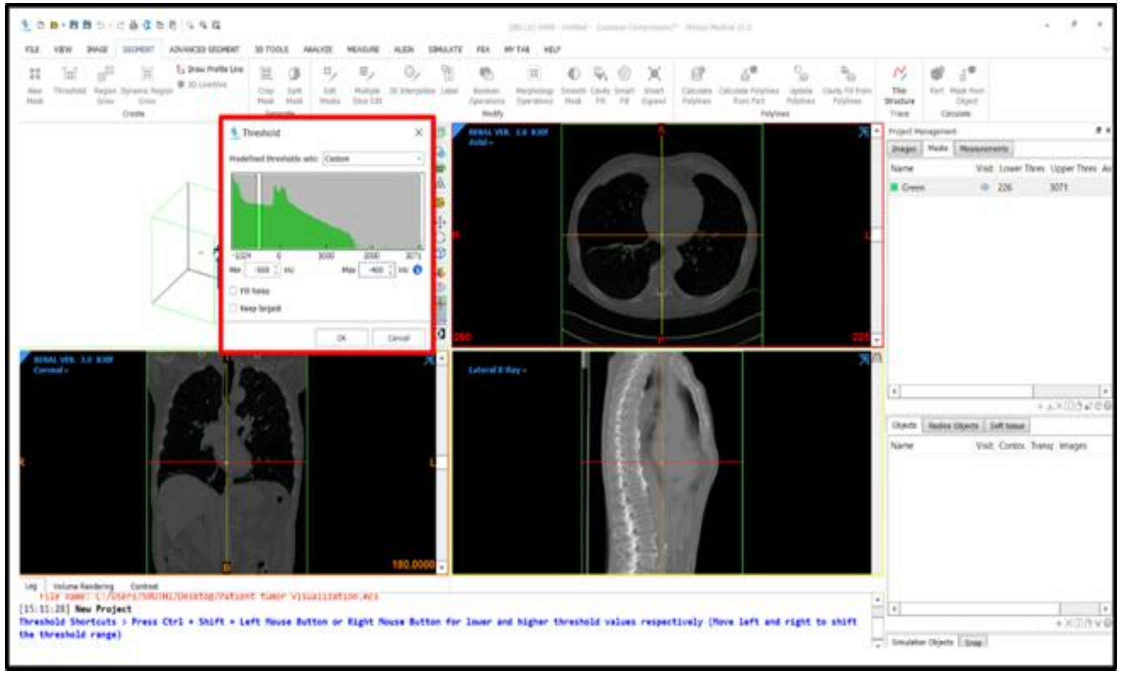


Fig 12: Applying threshold

Fig 13: Selecting the dynamic region grow

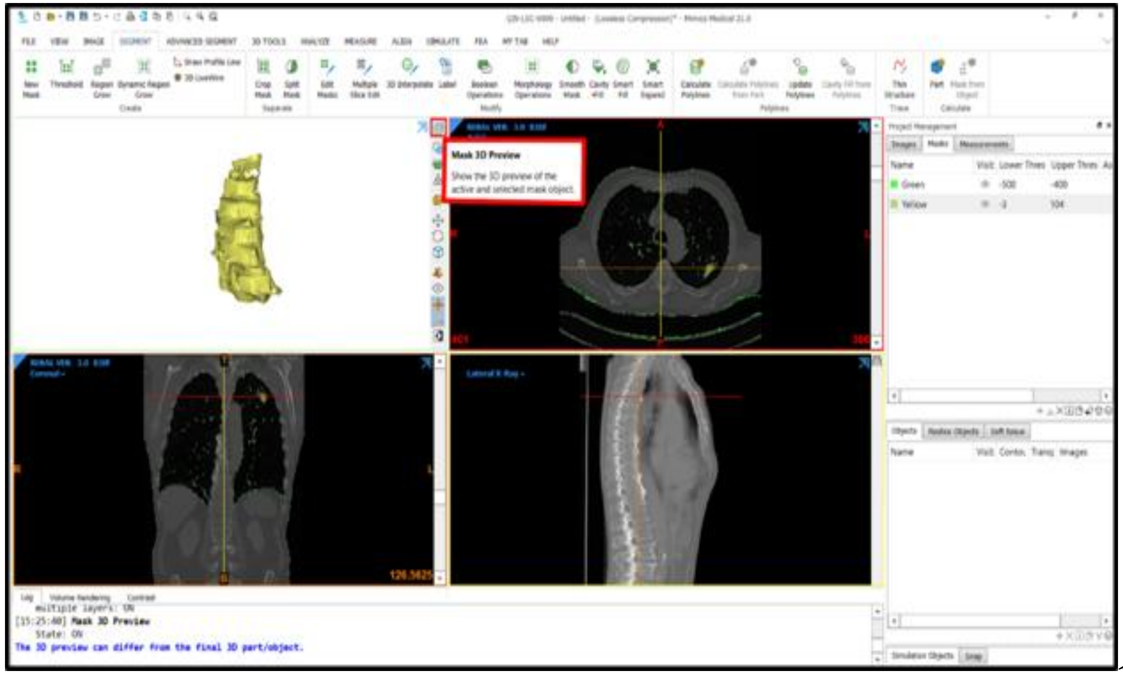
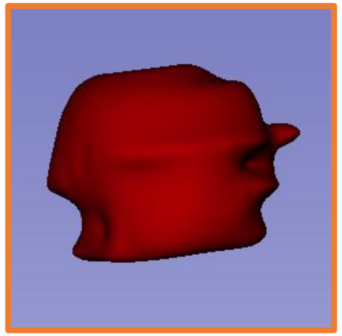


Fig 14: Three - Dimensional view of masked region

3D RECONSTRUCTION OF TUMOUR USING 3D SLICER SOFTWARE

- 3D Slicer software is used to three dimensionally reconstruct the tumour.
- At first, 2D stacked images such as CT or MRI are imported into the scene.
- 3D Slicer displays the image data in 27 different ways, each providing unique information.
- A desirable threshold, covering the region of interest is applied for masking.
- Each of the tumour slices are masked by using the paint tool.
- The Grow from Seeds tool is then selected. It implements the Grow Cut algorithm.
- The algorithm performs automated multi-label image segmentation using a set of user input scribbles.
- The ‘Show in 3D’ icon then displays the reconstructed tumour, whose surfaces can be smoothened using the smoothing factor tool.



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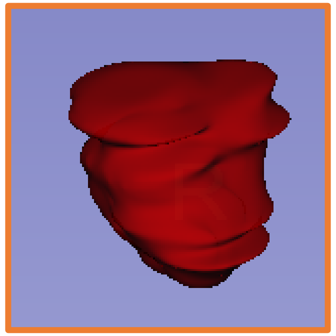

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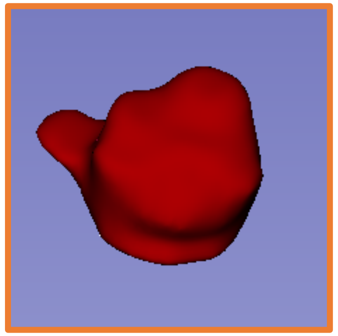

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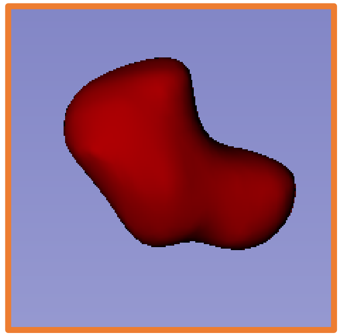

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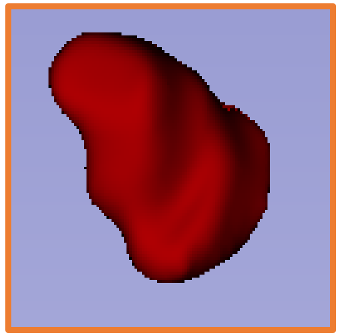

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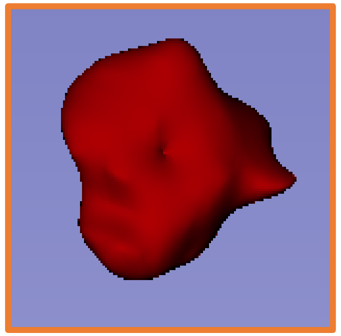

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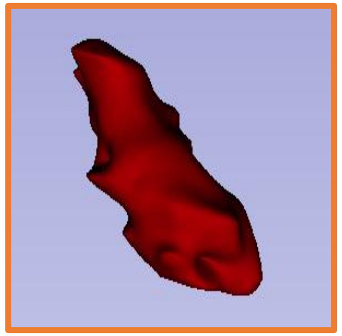

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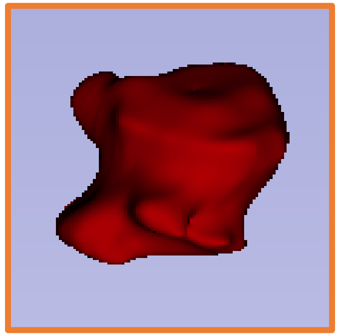

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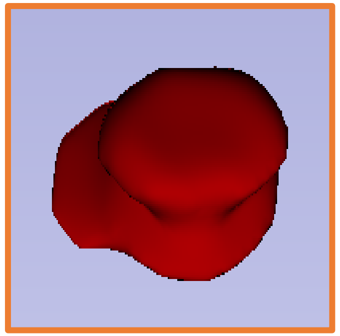


PATIENT 9

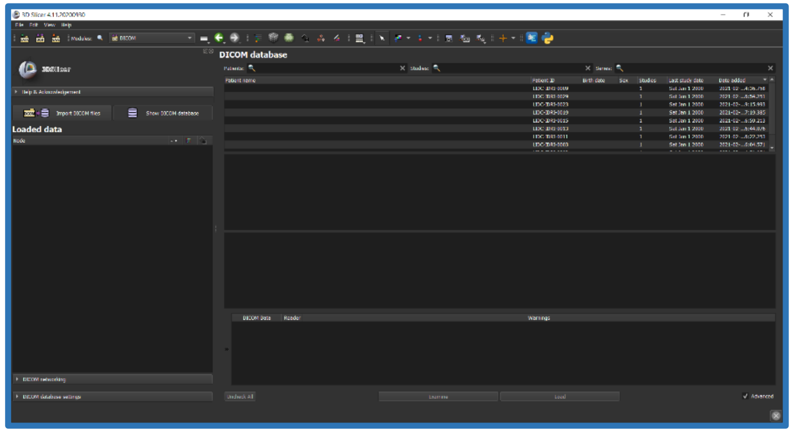


Fig 15: Loading of the dataset in 3D Slicer

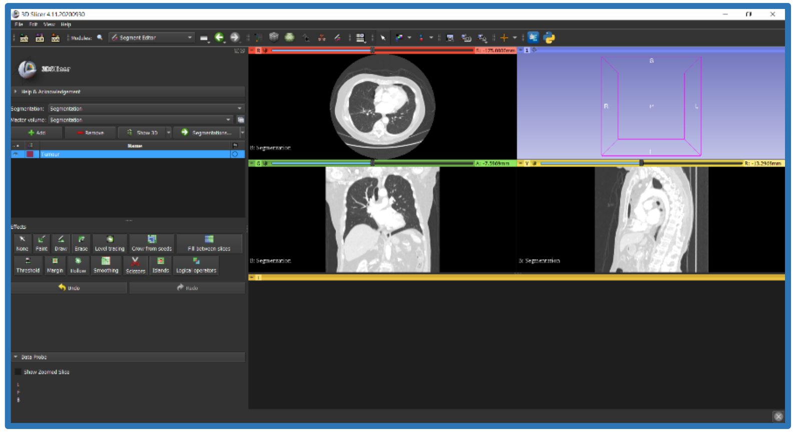


Fig 16: Segment Editor module in 3D Slicer

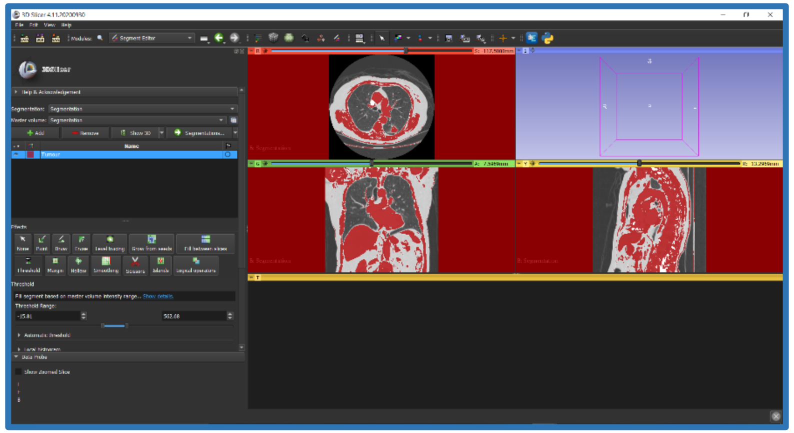


Fig 17: Applying a certain threshold

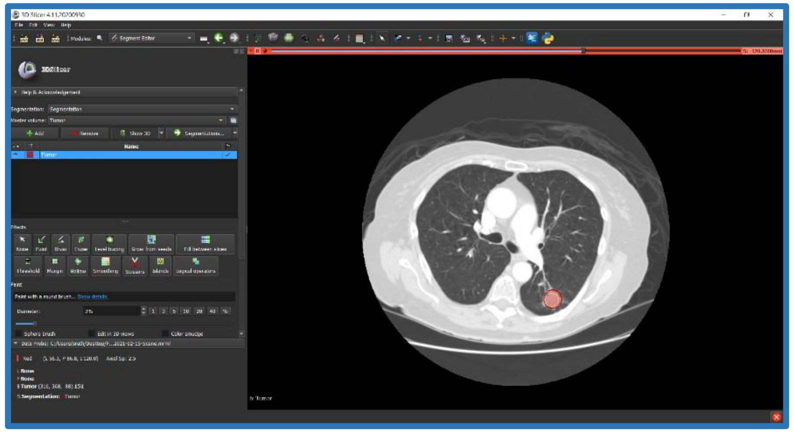


Fig 18: Painting a mask in each tumour slice

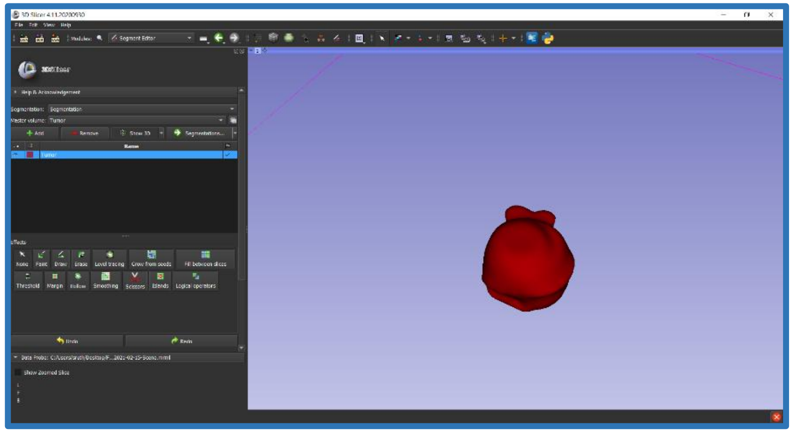


Fig 19: 3D reconstruction of tumour

COMPARISON OF TUMOUR VOLUMES

- Using the semi-automated algorithm, the tumour volumes are compared with clinically estimated gold standard values with an accuracy of 93%.
- In the Materialize MIMICS software, tumour volumes are slightly closer to clinically estimated Volume with an accuracy of 92%.
- In the 3D Slicer software, the estimated number of tumour volumes is higher than the clinically estimated tumour volumes is higher than the clinically estimated volume and produces an error rate of around 11.0%.
- The semi-automated algorithm gave the best estimation as it is curated using the annotation labels.
- Next, the Materialize MIMICS software is also an automated process that also holds good results.
- However, the 3D Slicer software doesn't give very much accurate results compared to others which are very really time-consuming, and most of the associated processes are manual.

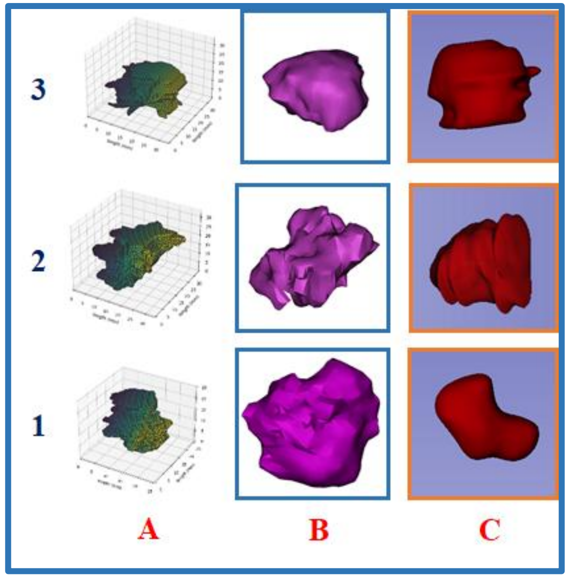


Fig 20: (A) 3D reconstructed tumour using algorithm (B) Materialize MIMICS software 3D reconstructed tumour (C) 3D Slicer software reconstructed tumour for patients 1, 2 and 3.

Table 2: p-value for various methods

Semi-Automated algorithm	Materialize Mimics	3D Slicer
0.04	0.07	0.3

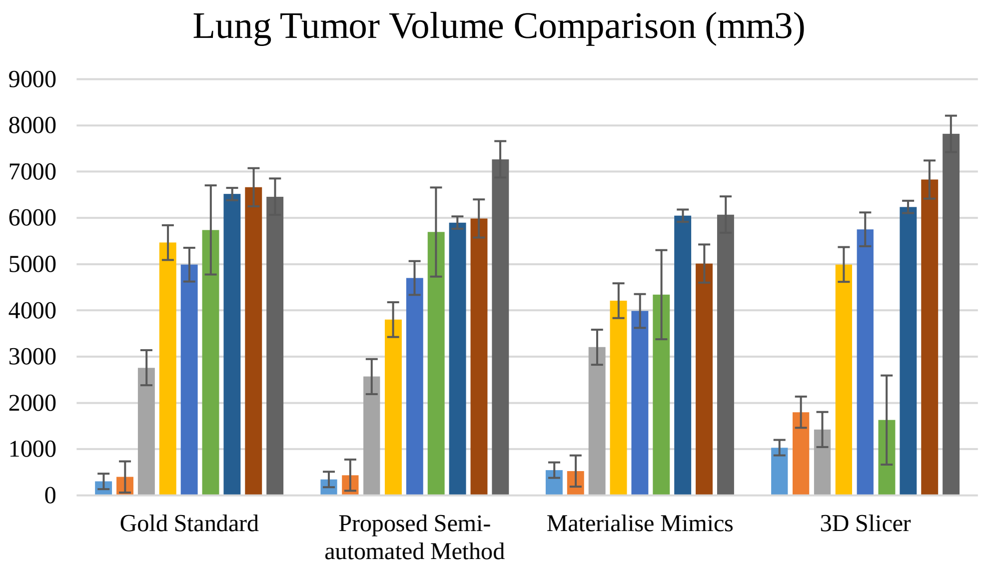


Fig 21: Statistical estimation of lung tumour using semi automate algorithms in comparison with Materialize Mimics and 3D Slicer.

Table 1: A comparison table of the various tumour volumes, surface areas and tumour dimensions

Patient Number	Tumour Volume (in mm3)	Tumour Surface Area (in mm2)	Tumour Dimensions (in mm)
#1	345.174	284.734	29.53 x 16.92
#2	438.415	369.353	28.38 x 15.68
#3	2569.1	1626.42	28.52 x 19.65
#4	3801.28	2341.8	29.89 x 36.88
#5	4701.3	1923	29.97 x 36.44
#6	5694.79	2816.42	23.48 x 30.64
#7	5899.56	2959.24	39.62 x 34.70
#8	5988.04	2880.76	23.34 x 32.16
#9	7267.64	2801.43	28.21 x 26.25

RADIOFREQUENCY ABLATION

- Radiofrequency ablation (RFA) is a minimally invasive, non-surgical procedure, that administers image guided electric needles through the skin to transmit radiofrequency waves to burn or ablate the tumour.
- The needles or probes are guided to the target site through imaging modalities such as Computed Tomography or Ultrasound. This technique is more efficient in localized tumours.
- This ablation technique is simulated using COMSOL Multiphysics software. COMSOL Multiphysics is a cross-platform finite element analysis, solver and multiphysics simulation software. It allows conventional physics-based user interfaces and coupled systems of partial differential equations (PDEs).
- The probe used for ablation is made of a trocar which is the main rod and electrode arm.
- The trocar base is electrically insulated. An electric current through the probe creates an electric field in the tissue. The field is strongest in the immediate vicinity of the probe and generates resistive heating, which dominates around the probe's electrode arms because of the strong electric field.
- While the original uses RF heating occurs with AC currents, the COMSOL Multiphysics model approximates the energy with DC currents.

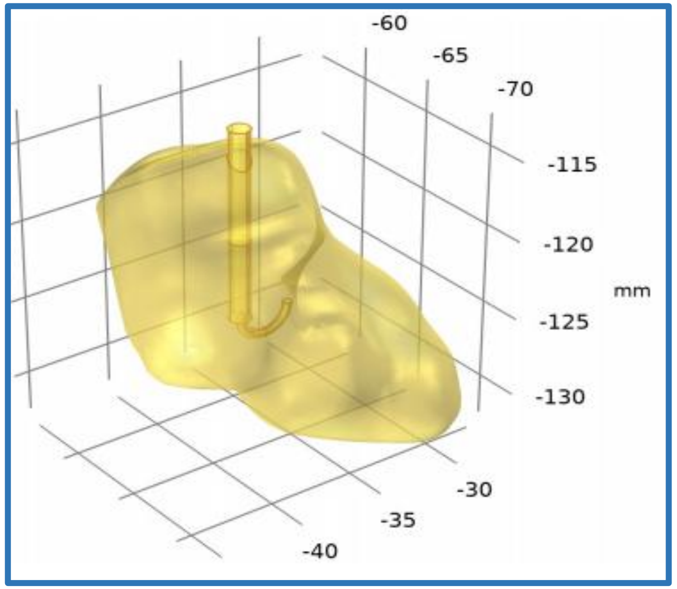


Fig 22: Trocar Probe in Lung Tumour

Table 3: Dimensions of Trocar Probe

	Radius (in mm)	Height (in mm)	Major Radius (in mm)	Minor Radius (in mm)
Trocar Base	0.5	12	-	-
Trocar Tip	0.5	3	-	-
Electrode	-	-	3	0.2667

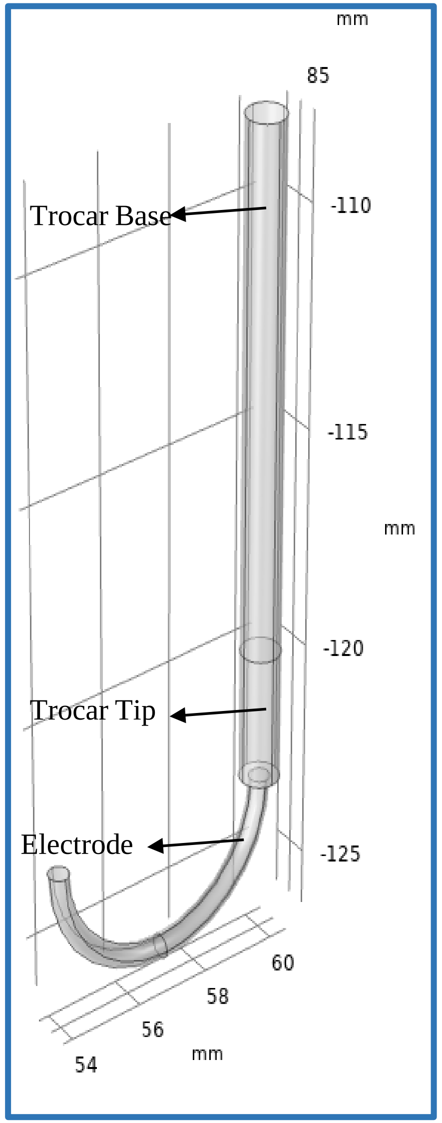


Fig 23: Trocar Probe

MULTIPHYSICS MODULE

- Two multiphysics modules are involved in Radiofrequency Ablation of lung tumour namely, Heat Transfer and Electric Current.
- The bioheat equation that governs the heat transfer in the tissue is as follows:

$$\delta_{ts}\rho C\frac{\partial T}{\partial t} + \nabla \cdot (-k\nabla T) = \rho_b C_b \omega_b (T_b - T) + Q_{met} + Q_{ext} \quad (4)$$

where δ_{ts} is a time-scaling coefficient; ρ is the tissue density; C is the tissue's specific heat; and k is its thermal conductivity. On the right side of the equation, ρ_b gives the blood's density; Cb is the blood's specific heat; ω_b is its perfusion rate; T_b is the arterial blood temperature; while Q_{met} and Q_{ext} are the heat sources from metabolism and spatial heating, respectively.

- In addition to the heat transfer equation this model provides a calculation of the tissue damage integral based on the Arrhenius equation:

$$\frac{\partial \alpha}{\partial t} = A\,\exp(-\frac{\Delta E}{RT}) \qquad (5)$$

where A is the frequency factor and ΔE is the activation energy for irreversible damage reaction

- These two parameters are dependent on the type of tissue. The fraction of necrotic tissue, θ_d, is then expressed by:

$$\theta_d = 1 - \exp(-\alpha) \qquad (6)$$

- The governing equation for the Electric Currents interface is:

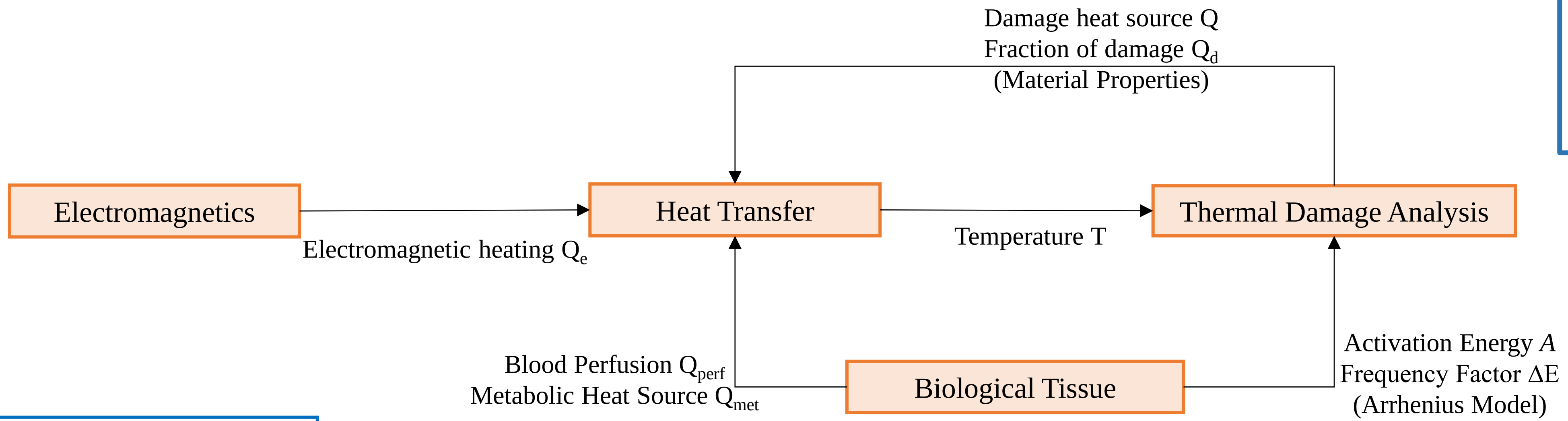
$$-\nabla \cdot (\sigma \nabla V - J^e) = Q_j \qquad (7)$$

- The simplified equation is given by:

$$-\nabla \cdot (\sigma \nabla V) = 0 \qquad (8)$$

- The boundary conditions at the tumour outer boundaries is ground, 0 V potential. At the electrode boundaries the potential equals 22 V. Assuming continuity for all other boundaries

- Finite element mesh realizes the physics of the given system and the characterization of its entire geometry.
- It reduces a complicated geometry to finite number of elements with simple geometries. It enables simulations with high levels of accuracy.
- Thus the multiphysics module can be summarized as follows:



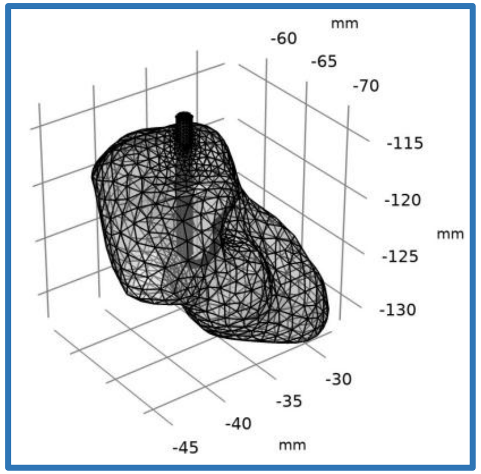


Fig 24: Applied Mesh to entire Geometry

Table 4: Material Properties

Material Property	Electrical conductivity (S/m)	Relative permittivity	Density (kg/m³)	Heat capacity at constant pressure (J/(kg·K))	Thermal conductivity (W/(m·K))	Frequency factor (1/s)	Activation energy (J/mol)
Lung Tumour	0.5	1	394	3886	0.39	1.67×10^{280}	1.71×10^{-6}
Lung Lobe	0.122	1	394	3886	0.39	1.67×10^{280}	1.71×10^{-6}
Electrode	1.00E+08	1	6450	840	18	-	-
Trocar Base	1.00E-05	1	70	1045	0.026	-	-
Trocar Tip	4.00E+06	1	21500	132	71	-	-

- The initial temperature of 36℃ is given through the probe.
- Figure 25 illustrates the voltage distribution across the electrode and tumour regions.
- Figure 26 visualizes the region where cancer cells die. The temperature has reached at least 50℃ . The isosurface for that temperature after 10 minutes is depicted.
- Figure 27 depicts how the temperature increases with time in the tissue around the electrode.
- The visualization of the fraction of necrotic tissue in the slice plot shown in Figure 28.
- From the fraction of damage graph, it is inferred that when passing minimum temperature of 36℃ for 10 minutes the fraction of necrotic tissue death is 1
- Probes are placed at the electrode tip, and at a distance of 1mm, 2mm and 3mm from the tip. It is used to represents the temperature distribution in the tumour tissue, at the tip of electrode and on above mentioned points, for different values of target temperature, when the voltage is applied for 10 minutes
- The Temperature Vs Time plot for each of these probes is plotted. The temperature at these points become constant after 30 seconds.

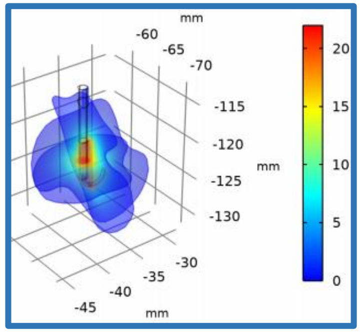


Fig 25: Multislice Electric Potential (V)

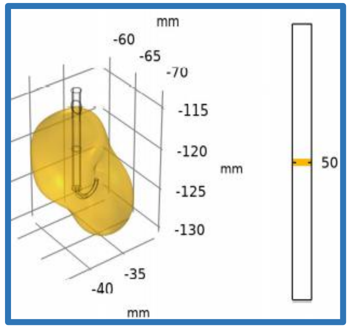


Fig 26: Isosurface Temperature (℃)

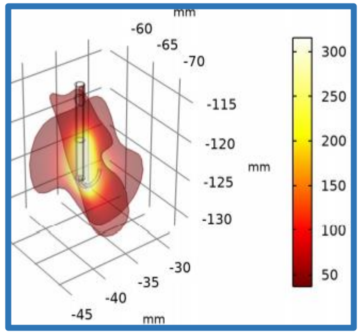


Fig 27: Slice Temperature (℃)

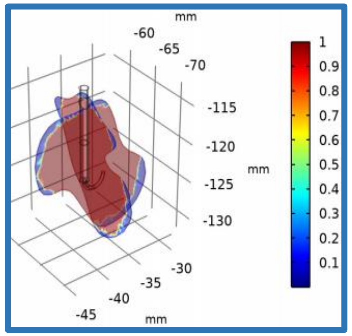


Fig 28: Fraction of Damage

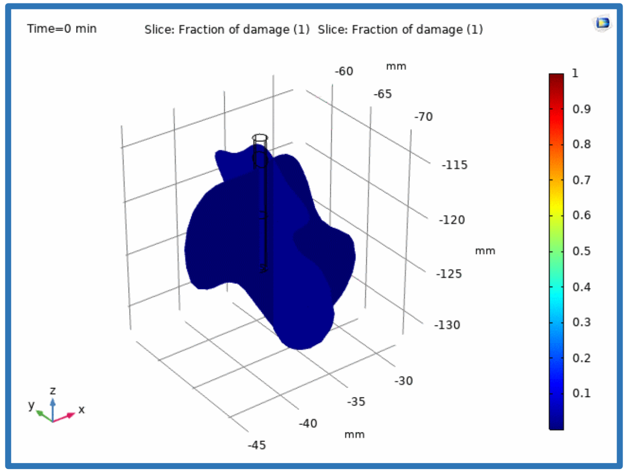


Fig 29: Fraction of Damage over time

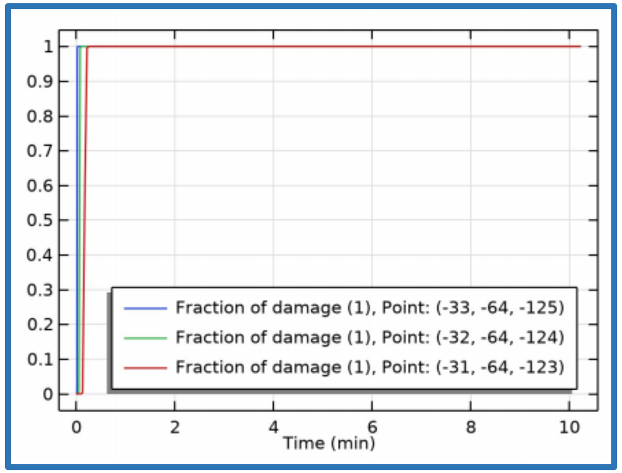


Fig 30: Fraction of damage graph

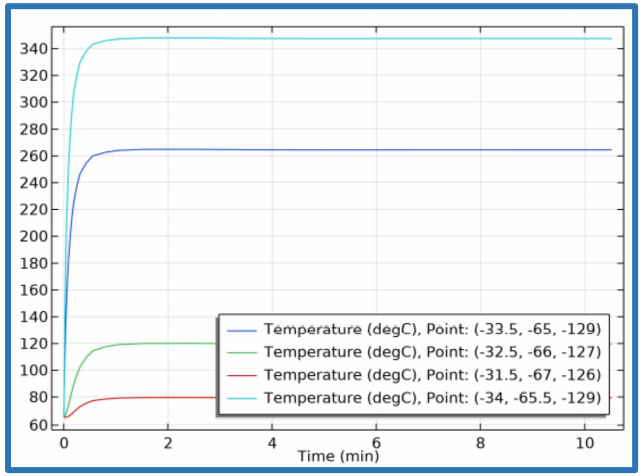


Fig 31: Temperature vs Time

PATIENT #2

Fig 32: Geometry of lung with tumour

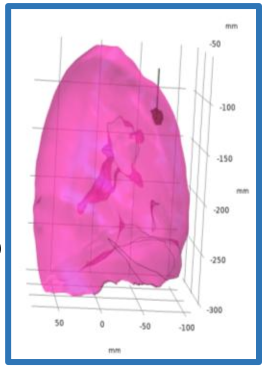


Fig 33: Finite element mesh for geometry

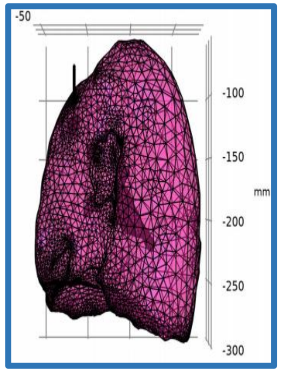


Fig 34: Temperature Distribution

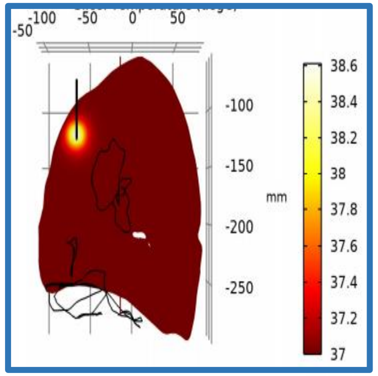
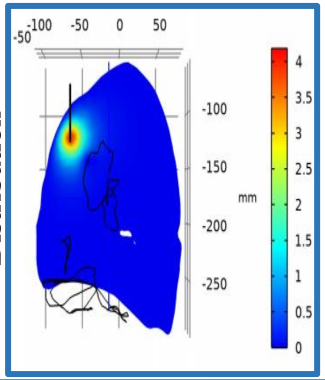
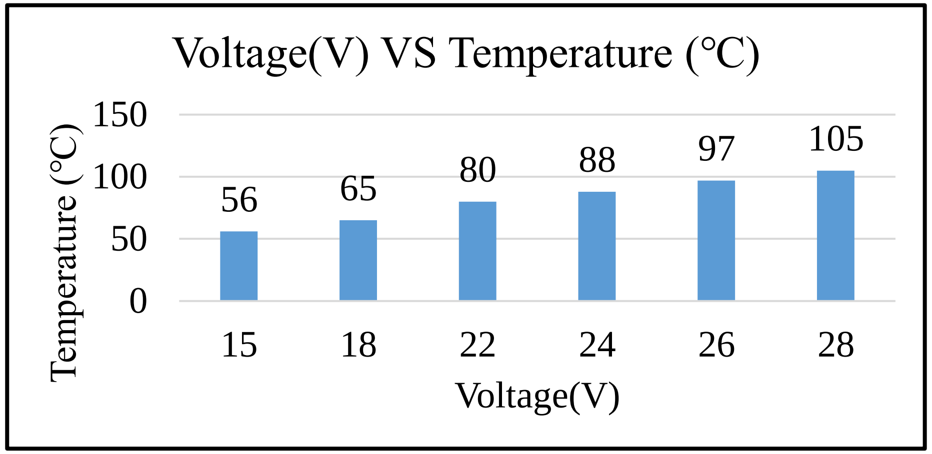


Fig 35: Voltage Distribution





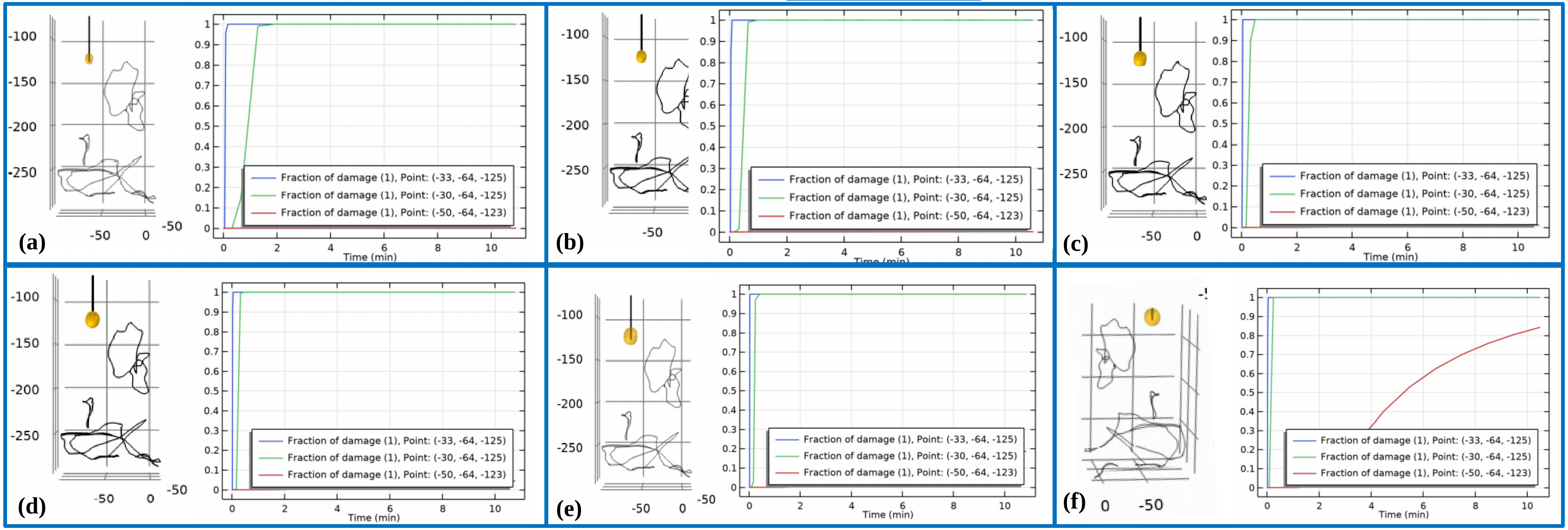


Fig 36: The applied voltage is (a) 15V (b) 18V (c) 22V (d) 24V (e) 26V and (f) 28V

PATIENT #8

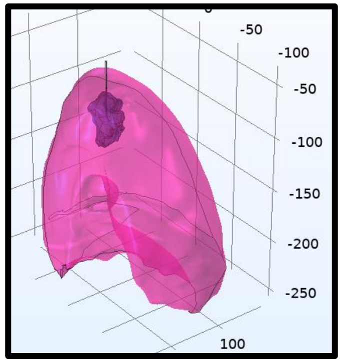


Fig 37: Geometry of lung with tumour

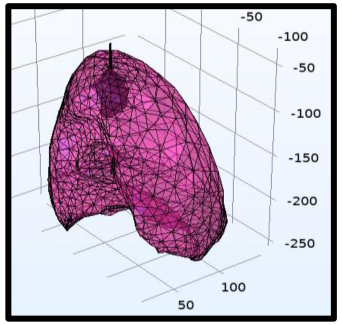


Fig 38: Finite element mesh for geometry

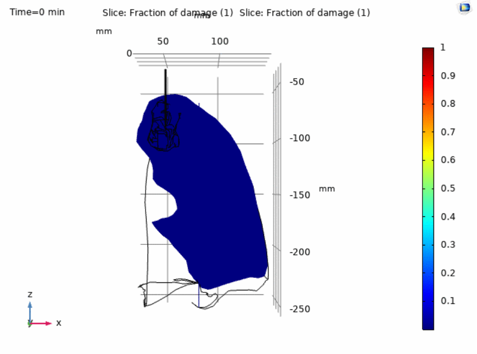


Fig 39: Fraction of Damage over time

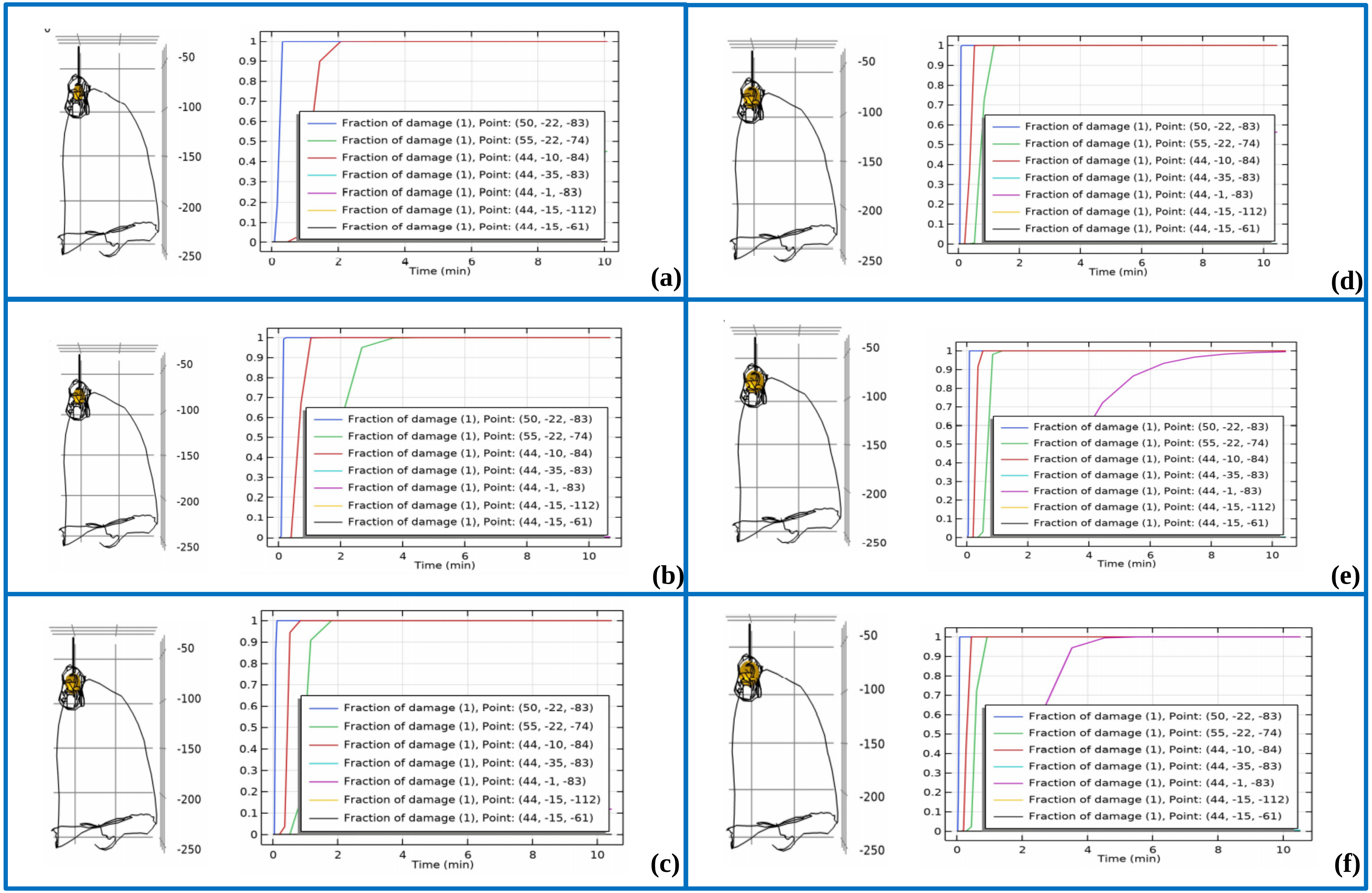


Fig 40: The applied voltage is (a) 15V (b) 18V (c) 22V (d) 24V (e) 26V and (f) 28V

PATIENT #5

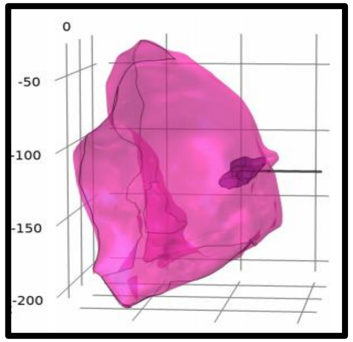


Fig 41: Geometry of lung with tumour

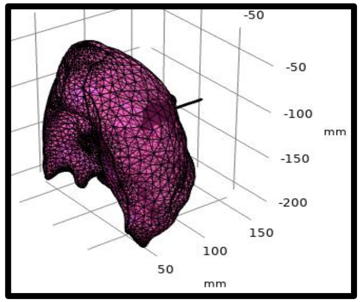


Fig 42: Finite element mesh for geometry

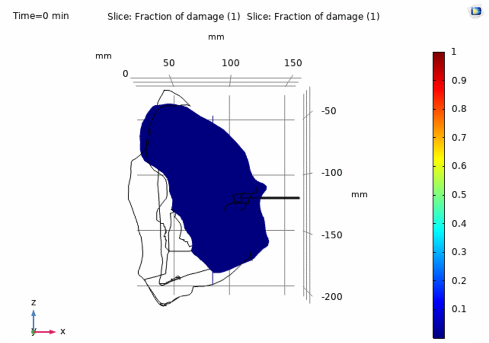


Fig 43: Fraction of Damage over time

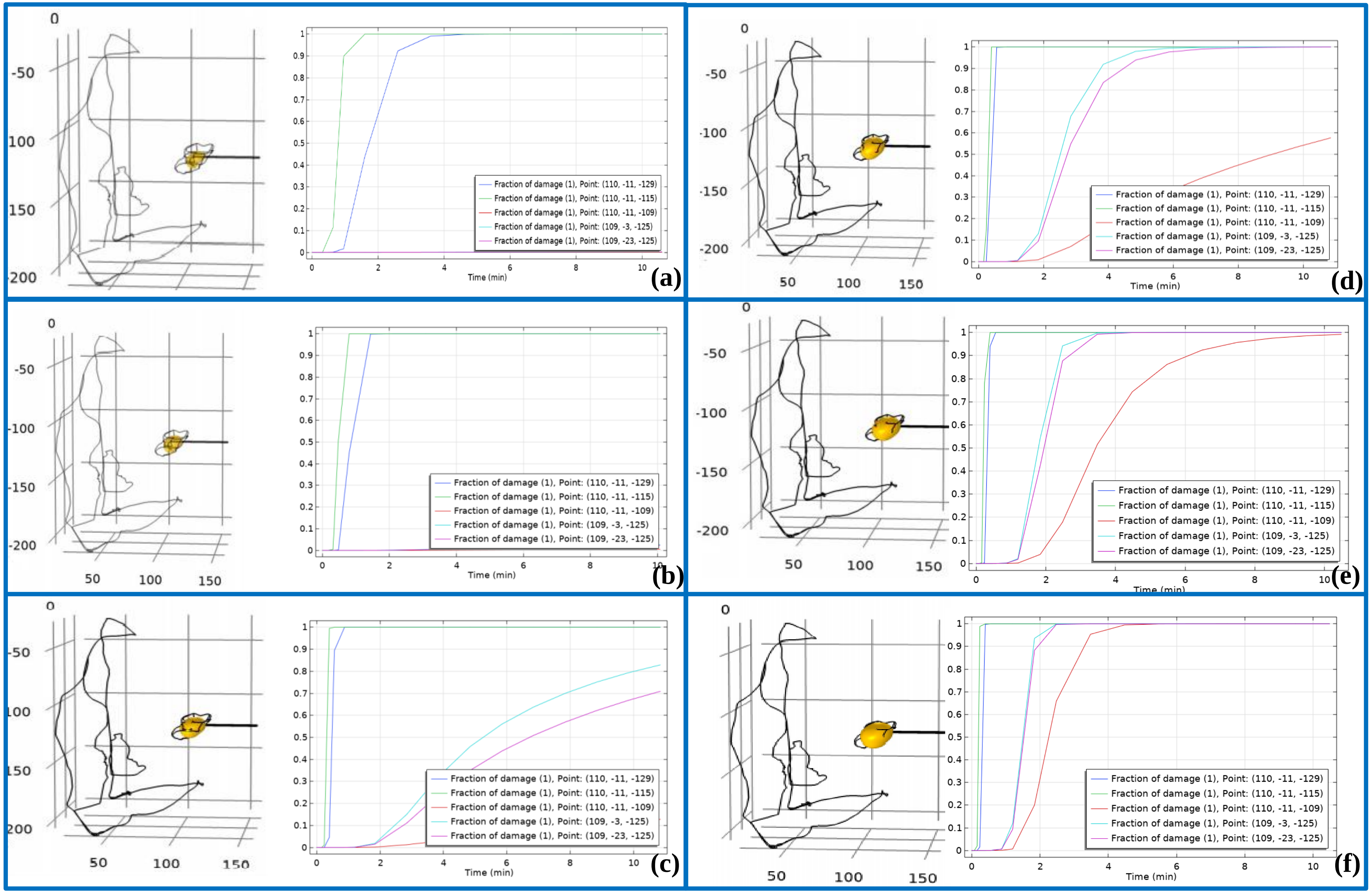


Fig 44: The applied voltage is (a) 15V (b) 18V (c) 22V (d) 24V (e) 26V and (f) 28V

INFERENCE

- 3D detection and visualization play a significant role in the improvisation of surgical planning for lung cancer treatment.
- The reconstruction techniques were compared with a developed semi-automated algorithm and software such as Materialize MIMICS and 3D Slicer.
- It was concluded from the comparison that Materialize MIMCS software output was the most accurate with the least error rate as it favors the research hypothesis.
- The Radiofrequency ablation modelling techniques and strategies for risk-free and successful thermal ablations of lung tumours was analyzed.
- The simulated analysis was done on a three-dimensional realistic human lung anatomy model with a tumour.
- The aim of this analysis was to determine the minimum target temperature and corresponding voltage for total tumor ablation during temperature regulated radiofrequency ablation of lung tumors.
- A temperature around 80°C that corresponds to 22V was sufficient to ablate the tumour with minimal damage to the surrounding healthy tissue.
- However, factors such as tumour size, tumour dimensions and tumour shape influence the ablation process.

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Thank You